

[Homework 2] Finite Markov Chains, Coupling

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Problem 1 (Optimal Coupling)

First recall the definition of $D_{TV}(\mu, \nu)$:

$$\begin{aligned} D_{TV}(\mu, \nu) &= \frac{1}{2} \sum_{i \in \Omega} |\mu(i) - \nu(i)| \\ &= \sum_{\substack{i \in \Omega, \\ \mu(i) \geq \nu(i)}} (\mu(i) - \nu(i)) = \sum_{\substack{i \in \Omega, \\ \mu(i) < \nu(i)}} (\nu(i) - \mu(i)) \end{aligned}$$

From the mathematical form, we can see that if we construct a coupling ω that satisfies

$$\Pr_{(X,Y) \sim \omega}[X = Y = i] = \omega(i, i) = \min\{\mu(i), \nu(i)\} \quad (1)$$

We get

$$\begin{aligned} \Pr_{(X,Y) \sim \omega}[X \neq Y] &= 1 - \sum_{i \in \Omega} \Pr_{(X,Y) \sim \omega}[X = Y = i] \\ &= \sum_{i \in \Omega} \mu(i) - \sum_{i \in \Omega} \min\{\mu(i), \nu(i)\} \\ &= \sum_{\substack{i \in \Omega, \\ \mu(i) \geq \nu(i)}} (\mu(i) - \nu(i)) \end{aligned}$$

Then we only have to prove that using equation (1) we can construct a legal coupling method.

We want to have a closed form solution for $\omega(i, j)$ when $i \neq j$,

$$\omega(i, j) = \frac{\max\{\mu(i) - \nu(i), 0\} \cdot \max\{\nu(j) - \mu(j), 0\}}{D_{TV}(\mu, \nu)} \quad (2)$$

I modified my expression based on lecture notes from MIT [2]. This construction is legal for

$$\begin{aligned} \sum_{j \in \Omega} \omega(i, j) &= \min\{\mu(i), \nu(i)\} + \sum_{\substack{j \in \Omega \\ j \neq i}} \omega(i, j) \\ &= \min\{\mu(i), \nu(i)\} + \max\{\mu(i) - \nu(i), 0\} \cdot \frac{\sum_{\substack{j \in \Omega \\ j \neq i}} \max\{\nu(j) - \mu(j), 0\}}{D_{TV}(\mu, \nu)} \end{aligned}$$

- when $\mu(i) \geq \nu(i)$

$$\begin{aligned} \sum_{j \in \Omega} \omega(i, j) &= \nu(i) + (\mu(i) - \nu(i)) \cdot \frac{\sum_{\substack{j \in \Omega \\ j \neq i}} (\nu(j) - \mu(j))}{D_{TV}(\mu, \nu)} \\ &= \nu(i) + \mu(i) - \nu(i) \\ &= \mu(i) \end{aligned}$$

- when $\mu(i) < \nu(i)$

$$\begin{aligned} \sum_{j \in \Omega} \omega(i, j) &= \mu(i) + 0 \cdot \frac{\sum_{\substack{j \in \Omega \\ j \neq i}} \max\{\nu(j) - \mu(j), 0\}}{D_{TV}(\mu, \nu)} \\ &= \mu(i) \end{aligned}$$

and it's similar to ν .

Problem 2 (Total Variation Distance is Non-Increasing)

We denote two random variables $X_t \sim \mu_t$, $Y_t \sim \pi$. We can construct a coupling method by

- X_t and Y_t move independently if they have not met each other.
- X_t and Y_t move together since they met.

This way,

$$\Delta(t+1) = \frac{1}{2} \sum_{i \in \Omega} |\mu_{t+1}(i) - \pi(i)| \leq \Pr[X^{t+1} \neq Y^{t+1}] \leq \Pr[X^t \neq Y^t] \quad (3)$$

In problem 1 we've proved that there exists a ω^* that satisfies

$$\Pr_{(X^t, Y^t) \sim \omega^*}[X^t \neq Y^t] = D_{TV}(\mu^t, \pi) = \Delta(t) \quad (4)$$

Thus with equation 3 and 4 we get $\Delta(t+1) \leq \Delta(t)$. We use a coupling composed of the one in problem 1 and the one mentioned above.

Problem 3

To make a simple guess: HT is more likely to appear first because it is more 'pointed'.

To explicitly describe the process, I draw the state map of sequence with length 2. There are in total 4 possible sequences with uniform distribution initially. From each state, we can obtain two possible states depending on the results of the coin tossing. We denote the minimum time a pattern X appears by T_X .

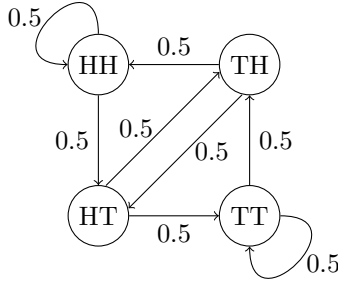


figure 1: Coin Toss Markov Chain with $n = 2$

We denote the minimum time to form a pattern Y from pattern X by $T_{X \rightarrow Y}$. Suppose we start at pattern HH , then it takes 0 to reach HH . If we start at HT , we have $\frac{1}{2}$ chance to get TH and then take $T_{TH \rightarrow HH}$ to get HH , and $\frac{1}{2}$ chance to get TT and then take $T_{TT \rightarrow HH}$ to get HH . It's similar to getting started at TH and TT . In total we have

$$\begin{cases} T_{HH \rightarrow HH} = 0 \\ T_{TH \rightarrow HH} = \frac{1}{2}(1 + T_{HT \rightarrow HH}) + \frac{1}{2}(1 + T_{HH \rightarrow HH}) = \frac{1}{2}T_{HT \rightarrow HH} + 1 \\ T_{HT \rightarrow HH} = \frac{1}{2}(1 + T_{TT \rightarrow HH}) + \frac{1}{2}(1 + T_{TH \rightarrow HH}) = \frac{1}{2}T_{HT \rightarrow HH} + \frac{1}{2}T_{TH \rightarrow HH} + 1 \\ T_{TT \rightarrow HH} = \frac{1}{2}(1 + T_{TT \rightarrow HH}) + \frac{1}{2}(1 + T_{TH \rightarrow HH}) = \frac{1}{2}T_{TT \rightarrow HH} + \frac{1}{2}T_{TH \rightarrow HH} + 1 \end{cases}$$

Consider the expectation,

$$\begin{cases} \mathbf{E}[T_{HH \rightarrow HH}] = 0 \\ \mathbf{E}[T_{TH \rightarrow HH}] = \frac{1}{2}\mathbf{E}[T_{HT \rightarrow HH}] + 1 \\ \mathbf{E}[T_{HT \rightarrow HH}] = \frac{1}{2}\mathbf{E}[T_{TT \rightarrow HH}] + \frac{1}{2}\mathbf{E}[T_{TH \rightarrow HH}] + 1 \\ \mathbf{E}[T_{TT \rightarrow HH}] = \frac{1}{2}\mathbf{E}[T_{TT \rightarrow HH}] + \frac{1}{2}\mathbf{E}[T_{TH \rightarrow HH}] + 1 \end{cases}$$

solve the system of equations, we get

$$\begin{cases} \mathbf{E}[T_{HH \rightarrow HH}] = 0 \\ \mathbf{E}[T_{TH \rightarrow HH}] = 4 \\ \mathbf{E}[T_{HT \rightarrow HH}] = 6 \\ \mathbf{E}[T_{TT \rightarrow HH}] = 6 \end{cases}$$

Then $\mathbf{E}[T_1] = 2 + \frac{1}{4}(0 + 4 + 6 + 6) = 6$. Similarly, $\mathbf{E}[T_2] = 2 + \frac{1}{4}(2 + 2 + 0 + 4) = 4$. The constant 2 refers to the time to get a certain initial state.

- **bonus:** To generalize, for a pattern with length n , we can simply use a state map with 2^n initial state with uniform distribution. According to the state map we can list the system of equations and solve them alike.

Problem 4 (Path Coupling)

At first I didn't really understand this question. I guess the statement of the problem skips too much detail. In class we designed a coupling for two random variables X and Y from μ and π respectively to show the mixing time can be bounded. However, here it says ' $\mathbf{E}[d(X_{t+1}, Y_{t+1}) \mid (X_t, Y_t)] \leq (1 - \alpha) \cdot d(X_t, Y_t)$ for those (X_t, Y_t) with $d(X_t, Y_t) = 1$ ', but in the coupling in class we had no such restrictions as $d(X_t, Y_t) = 1$. I later refer to more notes [3, 4] to find that $d(X_t, Y_t) = 1$ is not required in the example in class. It's something else we claim when constructing path coupling for sample coloring.

Now we verify about the irreducible component and its stationary distribution. When we try to transform a proper coloring to another proper coloring transform the by changing their disagreeing vertices one by one, if we met a improper intermediate state, we undo the most recent recoloring and seek a legal color instead to recolor and skip this vertex for later. We only need $q \geq \Delta + 2$ to ensure we can always find this replacement color. Repeat this process we can always reach the target state, and so this component is irreducible. Since we do not allow transition from a proper coloring to a improper one while improper coloring can move to a proper one, a improper coloring super-vertex has only out-degree but 0 in-degree. This indicates they are transient and have probability 0 in the stationary distribution. Thus the stationary distribution is still the uniform distribution on all proper coloring methods.

- (1) When we couple a pair of variables, we are actually making both decision depending on a same tossed coin. Now suppose we have coupled (Z_0, Z_1) to get (Z'_0, Z'_1) , Z'_0 is obtained by Z_0 taking one step on the Markov chain. The step taken by Z_1 depends on the high dimensional coin (a dice, let's say) tossed by Z_0 . As we go on to couple (Z_i, Z_{i+1}) , we are still making decision on this coin, and thus if viewed independently, we can say that z_0 and z_k moves independently. We claim $(X_{t+1}, Y_{t+1}) = (z_0, z_k)$ is a legal coupling of (X_t, Y_t) .

Here there is a confusing point. In the Markov chain mentioned in class, we do not allow any transition to improper coloring. Here there exists a coupling (defined in the third question) that holds the same rule on the new state space (i.e. we do not allow transitions that cause more improper pairs). I suggest in the first two question we assume the coupling is given and is an abstract one that satisfies $\mathbf{E}[d(X_{t+1}, Y_{t+1}) \mid (X_t, Y_t)] \leq (1 - \alpha) \cdot d(X_t, Y_t)$.

[Additional] According to the explanation given in class, I decide to provide a more explicit solution. Now suppose we have a path with length 3, i.e. Z_i, Z_{i+1} and Z_{i+2} . We have a legal couple for both pair with distance 1. Imagine we still have a coin here, and we already know the joint distribution. Suppose Z'_k have n_k possible destinations here. From the joint

distribution we can write

$$\Pr[Z_{i+1} = d_{i+1}(j)] = \sum_{k=1}^{n_i} \Pr_{(Z_i, Z_{i+1}) \sim \omega} [Z_{i+1} = d_{i+1}(j) \wedge Z_i = d_i(k)] \quad (5)$$

$$\Pr[Z_{i+2} = d_{i+2}(j)] = \sum_{k=1}^{n_{i+1}} \Pr_{(Z_{i+1}, Z_{i+2}) \sim \omega} [Z_{i+2} = d_{i+2}(j) \wedge Z_{i+1} = d_{i+1}(k)] \quad (6)$$

where $d_i(j)$ denotes the j -th destination of Z_i . Note that we couple Z_{i+2} conditioned on $Z'_{i+1} = z_{i+1}$, and I simplified the notation. With equation 5 and 6, we have

$$\begin{aligned} & \Pr[Z_{i+2} = d_{i+2}(j)]_{(Z_i, Z_{i+1}) \sim \omega \wedge (Z_{i+1}, Z_{i+2}) \sim \omega} \\ &= \sum_{k=1}^{n_{i+1}} \Pr_{(Z_{i+1}, Z_{i+2}) \sim \omega} [Z_{i+2} = d_{i+2}(j) \wedge Z_{i+1} = d_{i+1}(k)] \\ &= \sum_{k=1}^{n_{i+1}} \Pr_{\omega} [Z_{i+2} = d_{i+2}(j) \mid Z_{i+1} = d_{i+1}(k)] \cdot \Pr[Z_{i+1} = d_{i+1}(k)] \\ &= \sum_{k=1}^{n_{i+1}} \Pr_{\omega} [Z_{i+2} = d_{i+2}(j) \mid Z_{i+1} = d_{i+1}(k)] \cdot \sum_{l=1}^{n_i} \Pr_{\omega} [Z_{i+1} = d_{i+1}(k) \wedge Z_i = d_i(l)] \\ &= \sum_{k=1}^{n_{i+1}} \Pr_{\omega} [Z_{i+2} = d_{i+2}(j) \mid Z_{i+1} = d_{i+1}(k)] \cdot \sum_{l=1}^{n_i} \Pr_{\omega} [Z_{i+1} = d_{i+1}(k) \mid Z_i = d_i(l)] \cdot \Pr[Z_i = d_i(l)] \\ &= \sum_{l=1}^{n_i} \Pr[Z_{i+2} = d_{i+2}(j) \mid Z_i = d_i(l)] \cdot \Pr[Z_i = d_i(l)] \\ &= \sum_{l=1}^{n_i} \Pr[Z_{i+2} = d_{i+2}(j)] \end{aligned} \quad (7)$$

Now we knew the coupling is legal. If now we view Z_i and Z_{i+2} as a pair of adjacent couple, we can extend the legality to any length.

(2) In the coupling above, we ensure

$$\mathbf{E} [d(Z'_i, Z'_{i+1}) \mid (Z_i, Z_{i+1})] \leq (1 - \alpha) \cdot d(Z_i, Z_{i+1}) \quad (8)$$

Then,

$$\begin{aligned} \mathbf{E} [d(z_0, z_k) \mid (X, Y)] &\leq \mathbf{E} \left[\sum_{i=0}^{k-1} d(Z'_i, Z'_{i+1}) \mid (X, Y) \right] \\ &\leq (1 - \alpha) \sum_{i=0}^{k-1} d(Z_i, Z_{i+1}) \\ &= (1 - \alpha) \cdot k \end{aligned} \quad (9)$$

Note that Z_i is determined according to X and Y . We can replace the condition of conditional probability by (X, Y) .

(3) The path coupling implies $\mathbf{E} [d(X_t, Y_t) \mid (X_0, Y_0)] \leq (1 - \alpha)^t \cdot k$. We also know two coloring can be different at at most n vertices. Similar to the analysis of the sampling color in class, we have

$$\mathbf{E} [d(X_t, Y_t) \mid (X_0, Y_0)] \leq (1 - \alpha)^t \cdot n \leq n \cdot e^{-\alpha t} \quad (10)$$

If we want $\mathbf{E} [d(X_t, Y_t) \mid (X_0, Y_0)] \leq \epsilon$, we have a bound of mixing time

$$n \cdot e^{-\alpha t} \leq \epsilon, \tau_{mix}(\epsilon) \leq \frac{1}{\alpha} \log \frac{n}{\epsilon} \quad (11)$$

It seems we still have to perform a closed form solution of α by giving the explicit coupling method. Previously I was too young and simple. My coupling of two neighbor coloring can be described as follows.

For two arbitrary coloring X and Y in the expended state space satisfying $d(X, Y) = 1$, note that X and Y may not be proper coloring, we let v be the vertex that cause the difference. We use $X(v), Y(v)$ to denote the color of v in the two coloring. Now we uniformly select a vertex u and a color c .

- if $u = v$, recolor v with c in X (or Y) on this vertex if the move is legal.
- u is v 's neighbor, recolor u in X with c if proper. If $c = X(v)$ (or $Y(v)$), recolor u in Y with $Y(v)$ (or $X(v)$) and otherwise c if legal.
- u is not v 's neighbor, recolor v with c in X (or Y) on this vertex if legal.

The coupling is valid for both transition satisfy the rule, viewed in isolation. The third case will not change the distance for vertex u and its neighbors are the same in both states. The move is either legal for both or illegal for both. In the first case, the distance decreases to 0 if for both the move is legal and otherwise remain 1, that is

$$\Pr[d(X', Y') = 0 \mid u = v] \leq \frac{q - \Delta}{q}, \Pr[u = v] = \frac{1}{n} \quad (12)$$

We call this a good move. Recall that q is the number of colors and n is the number of vertices. In the second case, we ensure the distance is at least 1 because we don't change v . If c is $X(c)$, we turn both move illegal. If c is $Y(c)$, the distance may increase to 2 if we are lucky. We can easily give an upper bound assuming the move is always legal,

$$\Pr[d(X', Y') = 2 \mid u \in N(v)] \leq \frac{1}{q}, \Pr[u \in N(v)] = \frac{\Delta}{n} \quad (13)$$

This is a bad move. $N(v)$ denotes the neighbor set of v .

$$\begin{aligned} \mathbf{E}[d(X', Y') \mid (X, Y)] &= 1 - \Pr[d(X', Y') = 0 \mid u = v]\Pr[u = v] \\ &\quad + \Pr[d(X', Y') = 2 \mid u \in N(v)]\Pr[u \in N(v)] \\ &= 1 - \frac{q - \Delta}{qn} + \frac{\Delta}{qn} = 1 - \frac{q - 2\Delta}{qn} \leq 1 - \frac{1}{qn} \end{aligned} \quad (14)$$

Now we know $\alpha = \frac{1}{qn}$, $\tau_{mix}(\epsilon) \leq qn \log \frac{n}{\epsilon}$, i.e. mixing time is $O(n \log n)$.

References

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